Transformer Calculations

Exam code: 0625 & 0972

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Transformer calculations

- The voltages across the primary and secondary coils of a transformer can be calculated using the **transformer equation**, which states

The ratio of the voltages across the primary and secondary coils of a transformer is equal to the ratio of the number of turns on each coil

- It can be expressed by the equation:

$$\frac{\text{primary voltage}}{\text{secondary voltage}} = \frac{\text{number of turns on primary}}{\text{number of turns on secondary}}$$

- It can be expressed in symbols as follows:

$$\frac{V_p}{V_s} = \frac{N_p}{N_s}$$

- Where
  - V_p = voltage across the primary coil, in volts (V)
  - V_s = voltage across the secondary coil, in volts (V)
  - N_p = number of turns on the primary coil
  - N_s = number of turns on the secondary coil
- The transformer equation can be flipped upside down to give:

Help me understand

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Got it

- Rearranging for the secondary voltage:

$$V_s = V_p \times \frac{N_s}{N_p}$$

- This equation shows that the output (**secondary**) voltage of a transformer depends on:
  - the **number of turns** on the primary and secondary coils
  - the input (**primary**) voltage
- In a step-up transformer, $V_s > V_p$ and $N_s > N_p$
- In a step-down transformer, $V_s < V_p$ and $N_s < N_p$

Worked Example

A transformer has 20 turns on the primary coil and 800 turns on the secondary coil. The voltage across the primary coil is 500 V.

a) Calculate the output voltage of the secondary coil.

b) State whether this is a step-up or step-down transformer.

Answer

Part (a)

Step 1: List the known quantities

- Number of turns on the primary coil, N_p = 20
- Number of turns on the secondary coil, N_s = 800
- Voltage across the primary coil, V_P = 500 V

Step 2: Write down the transformer equation

- There will be less rearranging to do if V_S is on the top of the fraction

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

Step 3: Rearrange the equation to make V_S the subject

$$V_s = V_p \times \frac{N_s}{N_p}$$

Step 4: Substitute the known values into the equation

$$V_S = 500 \times \frac{800}{20} = 20\,000 \text{ V}$$

Part (b)

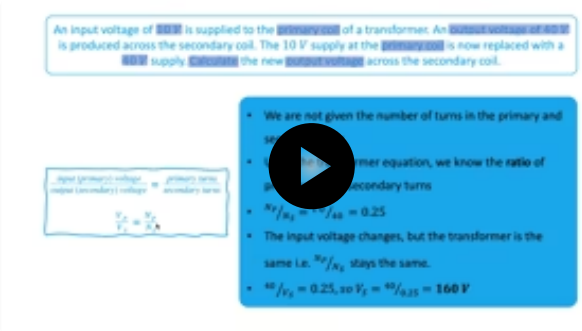
- The secondary voltage is larger than the primary, $V_s > V_p$
- There are more turns on the secondary coil than on the primary, $N_s > N_p$

Examiner Tips and Tricks

When carrying out transformer calculations, make sure you have used the **same letter** (p or s) in the **numerators** (top line) of the fraction and the **same letter** (p or s) in the **denominators** (bottom line) of the fraction.

There will be less rearranging to do in a calculation if the variable which you are trying to find is on the numerator (top line) of the fraction.

The individual loops of wire going around each side of the transformer should be referred to as **turns** and **not** coils.



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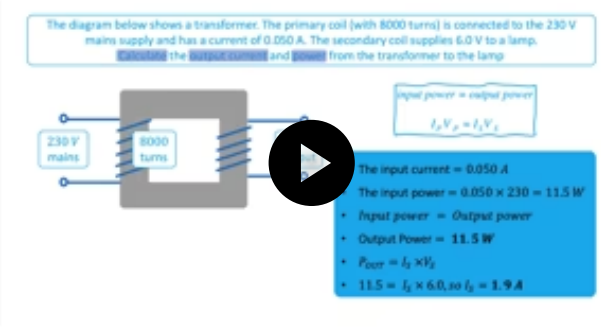

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Ideal transformer equation

Extended tier only

power source, due to the **law of conservation of energy**, they cannot increase the power output

- If a transformer is 100% efficient, then the input power in the primary coil is equal to the output power of the secondary coil:

$$\text{power in primary} = \text{power in secondary}$$

- The equation to calculate electrical power is:

$$P = V \times I$$

- Where:
  - P = power, in watts (W)
  - V = voltage, in volts (V)
  - I = current, in amps (A)

- Therefore, the equation for an ideal transformer is:

$$I_p V_p = I_s V_s$$

- Where:
  - I_p = primary current, in amps (A)
  - V_p = primary voltage, in volts (V)
  - I_s = secondary current, in amps (A)
  - V_s = secondary voltage, in volts (V)

- The equation above could also be written as a ratio:

$$\frac{I_s}{I_p} = \frac{V_p}{V_s}$$

Worked Example

A transformer in a travel adapter steps up a 115 V a.c. mains electricity supply to the 230 V needed for a hair dryer. A current of 5 A flows through the hairdryer.

Assuming that the transformer is 100% efficient, calculate the current drawn from the mains supply.

Answer:

Step 1: List the known quantities

- Voltage in primary coil, V_p = 115 V
- Voltage in secondary coil, V_s = 230 V
- Current in secondary coil, I_s = 5 A

Step 2: Write down the equation for an ideal transformer

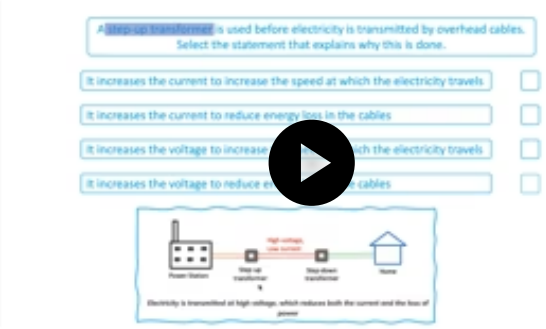
$$V_p \times I_p = V_s \times I_s$$

Step 3: Substitute in the known values

$$115 \times I_p = 230 \times 5$$

Step 4: Rearrange the equation to find the primary current

$I_p = 10\ \text{A}$



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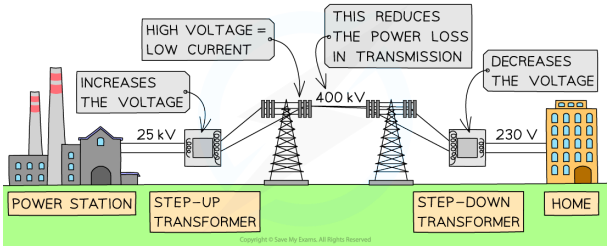
High-voltage transmission

- Electricity is transmitted through power cables at a low current to prevent dissipation of energy
  - When current flows in a wire, there is heating in the wire due to resistance
  - Therefore, energy is dissipated to the surroundings, this energy is wasted
  - The lower the current, the more efficient the energy transfer
- Electrical power is equal to voltage × current, or $P = VI$
- This means that a low current can be achieved by increasing the voltage, so electricity must be transmitted at a **high voltage**
  - A **smaller current** flowing through the power lines results in **less heat** being produced in the wire
  - This **reduces** the **energy loss** in the power lines
- The key **advantages** of high-voltage transmission of electricity are:
  - the reduced power loss in transmission cables increases the efficiency of energy transfer
  - lower currents in cables mean thinner, and therefore, cheaper cables can be used



Electricity is transmitted at high voltage, reducing the current and hence power loss in the cables using

Calculating power losses

Extended tier only

- The power dissipated in the wire due to resistance is given by:

$$P = I^2R$$

- Where:
  - P = power, in watts (W)
  - I = current, in amps (A)
  - R = resistance, in ohms (Ω)
- A **step-up** transformer is used to increase the voltage and decrease the current of electricity **before** transmission
  - A high-voltage transmission ensures the same power transfer with a smaller current
  - A smaller current means less thermal energy will be lost due to the resistance in the wire
- A **step-down** transformer is used to decrease the voltage and increase the current of electricity **after** transmission
  - High-voltage electricity is dangerous for use in homes, so it must be lowered before the current reaches consumers

Examiner Tips and Tricks

If you forget the equation $P = I^2R$ just remember 'Twinkle twinkle little star, power equals I squared R''.

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